

Aggregate functions:

Customers [-]

customer_id [int]

first_name [varchar(100)]

last_name [varchar(100)]

age [int]

country [varchar(100)]

Orders [-]

order_id [integer]

item [varchar(100)]

amount [integer]

customer_id [integer]

Shippings [-]

shipping_id [integer]

status [integer]

customer [integer]

Input

Step 1: Create Table

```
CREATE TABLE bank_transactions (
  txn_id INTEGER PRIMARY KEY,
  customer_name VARCHAR(50),
  branch_name VARCHAR(50),
  transaction_type VARCHAR(20),
  amount DECIMAL(10,2),
  transaction_date DATE
);
```

Step 2: Insert Data

```
INSERT INTO bank_transactions VALUES
```

Run SQL

Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
1	Pending	2
2	Pending	4
3	Delivered	3
4	Pending	5
5	Delivered	1

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

Step 20: GROUP BY + AVG()

```
SELECT branch_name,
  AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Step 21: WHERE + GROUP BY + HAVING

```
SELECT branch_name,
  SUM(amount) AS Total_Deposit
FROM bank_transactions
```

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

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Customers [-]

customer_id [int]

first_name [varchar(100)]

last_name [varchar(100)]

age [int]

country [varchar(100)]

Orders [-]

order_id [integer]

item [varchar(100)]

amount [integer]

customer_id [integer]

Shippings [-]

shipping_id [integer]

status [integer]

customer [integer]

-- Step 20: GROUP BY + AVG()

SELECT branch_name,
AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name;

-- Step 21: WHERE + GROUP BY + HAVING

SELECT branch_name,
SUM(amount) AS Total_Deposit
FROM bank_transactions

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
1	Pending	2
2	Pending	4
3	Delivered	3

Customers [-]

customer_id [int]

first_name [varchar(100)]

last_name [varchar(100)]

age [int]

country [varchar(100)]

Orders [-]

order_id [integer]

item [varchar(100)]

amount [integer]

customer_id [integer]

Shippings [-]

shipping_id [integer]

status [integer]

customer [integer]

FROM bank_transactions
WHERE transaction_type = 'Deposit'
GROUP BY branch_name
HAVING SUM(amount) > 15000;

-- Step 22: GROUP BY + HAVING + ORDER BY

SELECT branch_name,
AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING AVG(amount) > 7000
ORDER BY Avg_Amount DESC;

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

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Customers [-]

customer_id [int]
first_name [varchar(100)]
last_name [varchar(100)]
age [int]
country [varchar(100)]

Orders [-]

order_id [integer]
item [varchar(100)]
amount [integer]
customer_id [integer]

Shippings [-]

shipping_id [integer]
status [integer]
customer [integer]

Input

```
FROM bank_transactions
WHERE transaction_type = 'Deposit'
GROUP BY branch_name
HAVING SUM(amount) > 15000;

-- Step 22: GROUP BY + HAVING + ORDER BY

SELECT branch_name,
       AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING AVG(amount) > 7000
ORDER BY Avg_Amount DESC;
```

Run SQL

Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
-------------	--------	----------

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

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Customers [-]

customer_id [int]
first_name [varchar(100)]
last_name [varchar(100)]
age [int]
country [varchar(100)]

Orders [-]

order_id [integer]
item [varchar(100)]
amount [integer]
customer_id [integer]

Shippings [-]

shipping_id [integer]
status [integer]
customer [integer]

Input

```
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) > 3;

-- Step 20: GROUP BY + AVG()

SELECT branch_name,
       AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name;

-- Step 21: WHERE + GROUP BY + HAVING
```

Run SQL

Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
-------------	--------	----------

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

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Customers [-]

customer_id [int]
first_name [varchar(100)]
last_name [varchar(100)]
age [int]
country [varchar(100)]

Orders [-]

order_id [integer]
item [varchar(100)]
amount [integer]
customer_id [integer]

Shippings [-]

shipping_id [integer]
status [integer]
customer [integer]

Input

-- Step 17: WHERE + ORDER BY

SELECT *
FROM bank_transactions
WHERE amount > 8000
ORDER BY amount DESC;

-- Step 18: GROUP BY + COUNT()

SELECT branch_name,
COUNT(*) AS Customer_Count
FROM bank_transactions
GROUP BY branch_name;

Run SQL

Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
-------------	--------	----------

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

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Customers [-]

customer_id [int]
first_name [varchar(100)]
last_name [varchar(100)]
age [int]
country [varchar(100)]

Orders [-]

order_id [integer]
item [varchar(100)]
amount [integer]
customer_id [integer]

Shippings [-]

shipping_id [integer]
status [integer]
customer [integer]

Input

SELECT branch_name,
MAX(amount) AS Highest_Amount
FROM bank_transactions
GROUP BY branch_name;

-- Step 16: GROUP BY + MIN()

SELECT branch_name,
MIN(amount) AS Lowest_Amount
FROM bank_transactions
GROUP BY branch_name;

Run SQL

Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
-------------	--------	----------

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

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Customers [-]

customer_id [int]

first_name [varchar(100)]

last_name [varchar(100)]

age [int]

country [varchar(100)]

Orders [-]

order_id [integer]

item [varchar(100)]

amount [integer]

customer_id [integer]

Shippings [-]

shipping_id [integer]

status [integer]

customer [integer]

Input

-- Step 13: WHERE + COUNT()

SELECT COUNT(*) AS Withdrawals
FROM bank_transactions
WHERE transaction_type = 'Withdrawal';

-- Step 14: WHERE + AVG()

SELECT AVG(amount) AS Avg_Deposit
FROM bank_transactions
WHERE transaction_type = 'Deposit';

Run SQL

Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
-------------	--------	----------

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

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Customers [-]

customer_id [int]

first_name [varchar(100)]

last_name [varchar(100)]

age [int]

country [varchar(100)]

Orders [-]

order_id [integer]

item [varchar(100)]

amount [integer]

customer_id [integer]

Shippings [-]

shipping_id [integer]

status [integer]

customer [integer]

Input

FROM bank_transactions
GROUP BY branch_name
HAVING SUM(amount) > 20000;

-- Step 11: GROUP BY + ORDER BY

SELECT branch_name,
SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
ORDER BY Total_Amount DESC;

Run SQL

Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
-------------	--------	----------

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

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Customers [-]

- customer_id [int]
- first_name [varchar(100)]
- last_name [varchar(100)]
- age [int]
- country [varchar(100)]

Orders [-]

- order_id [integer]
- item [varchar(100)]
- amount [integer]
- customer_id [integer]

Shippings [-]

- shipping_id [integer]
- status [integer]
- customer [integer]

Input

```
SELECT SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type = 'Deposit';

-- Step 9: GROUP BY + SUM()

SELECT branch_name,
SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Run SQL

Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
-------------	--------	----------

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

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Customers [-]

- customer_id [int]
- first_name [varchar(100)]
- last_name [varchar(100)]
- age [int]
- country [varchar(100)]

Orders [-]

- order_id [integer]
- item [varchar(100)]
- amount [integer]
- customer_id [integer]

Shippings [-]

- shipping_id [integer]
- status [integer]
- customer [integer]

Input

```
-- Step 6: MIN()

SELECT MIN(amount) AS Lowest_Transaction
FROM bank_transactions;

-- Step 7: COUNT()

SELECT COUNT(*) AS Total_Transactions
FROM bank_transactions;
```

Run SQL

Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
-------------	--------	----------

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

Customers [-]

customer_id [int]
first_name [varchar(100)]
last_name [varchar(100)]
age [int]
country [varchar(100)]

Orders [-]

order_id [integer]
item [varchar(100)]
amount [integer]
customer_id [integer]

Shippings [-]

shipping_id [integer]
status [integer]
customer [integer]

Input

-- Step 3: SUM()

```
SELECT SUM(amount) AS Total_Amount  
FROM bank_transactions;
```

-- Step 4: AVG()

```
SELECT AVG(amount) AS Average_Transaction  
FROM bank_transactions;
```

-- Step 5: MAX()

Output

first_name	age
John	31
Robert	22
David	22
John	25
Betty	28

Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	4
5	Mousepad	250	4

Shippings

shipping_id	status	customer
-------------	--------	----------